

ENVIRONMENT	2012	2018	2019
Energy			
Employee Numbers for Intensity Metrics			
Intensity-based employee FTEs - used for environmental intensity-based metrics*	N/A	7,787	7,600
Total Workers (including employees and contingent workers within our buildings, on FTE basis)- used for environmental intensity-based metrics*	N/A	8,763	8,422
Energy Use (MWh)			
Total direct and indirect energy use- real estate and transport (MWh)	153,210	114,822	105,699
Total direct and indirect energy use- real estate and transport (Billion Joules)	551,556	413,359	380,407
Total direct and indirect energy use- real estate only (MWh)**	148,266	111,469	102,438
Direct energy use- fossil fuels (MWh)	30,966	28,525	27,646
Indirect energy use- purchased electricity, steam, chilled water in metered buildings (MWh)	93,779	60,876	56,960
Indirect energy use- estimated electricity use in leased (and unmetered) buildings (MWh)	28,465	16,518	13,623
Energy Intensity			
Total direct and indirect energy use- real estate** (MWh) per square foot at year end	0.0267	0.0253	0.0253
Total direct and indirect energy use- real estate** (MWh) per square foot - average square feet over four quarters	N/A	0.0245	0.0234
Total direct and indirect energy use (MWh) per full time employee (FTE)	N/A	14.74	13.90

* Our Human Resources-generated FTE number uses a slightly different calculation than the Annual Report FTE number and allows us to break down workers by individual locations. This allows for intensity metrics to be generated at the facility level. We use the intensity-based FTE number for our real estate-related intensity-based metrics.

** Total direct and indirect energy use- real estate only includes Natural Gas, Diesel Fuel, Metered and Estimated Electricity, Estimated Natural Gas and Propane Gas for heat, Steam and Chilled Water. Steam and chilled water through 2015. Estimated Natural Gas and Propane Gas for heat were new in 2015.

ENVIRONMENT	2012	2018	2019
Climate Change and Greenhouse Gas (GHG) Emissions*			
Total GHG Emissions from Scopes 1 & 2 Real Estate and Scope 1 Travel (Metric tons of CO ₂ e)	81,735	46,287	42,621
GHG emissions- Scopes 1 and 2 Real Estate only- (Metric tons of CO ₂ e)	80,533	45,477	41,828
Emissions intensity- real estate (metric tons of CO ₂ e per square foot at year end) within organization	0.0145	0.0103	0.0096
Emissions intensity- real estate (metric tons of CO ₂ e per square foot-average square feet over four quarters) within organization	N/A	0.0100	0.0096
Percent Cumulative Scope 1 & 2 Real Estate GHG reduction from 2012 base year (Goal= 20 percent GHG reduction from base year by 2020)	N/A	43.5	48.1
GHG emissions- Scope 1 transport only	1,201	810	793
Breakdown by Scope			
Scope 1 GHG emissions- natural gas, jet fuel, diesel, gasoline, and refrigerants	6,950	6,371	6,304
Scope 2 GHG emissions- electricity, steam, and chilled water in company-controlled buildings	74,784	39,916	36,316
Scope 3 GHG Emissions (Metric tons of CO ₂ e)- Supply Chain and Transport*			
Purchased Goods and Services			
Total lifecycle paper emissions**	3,055	5,257	4,560
Life cycle emissions-office copy paper use**	1,270	1,964	1,662
Life cycle emissions-other office paper use**	1,563	2,930	2,721
Life cycle emissions-marketing paper use**	222	364	176
Total lifecycle computer emissions	520	3,031	2,816
Total lifecycle carpeting emissions	137	289	504

*Comerica utilizes the Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard (Revised Edition) for purposes of calculating its greenhouse gas (GHG) emissions. Specific methodologies for estimating Scope 1, Scope 2, and Scope 3 emissions, including activity data sources, estimation techniques, global warming potentials, and emission conversion factors, are disclosed in Comerica's most recent CDP Response.

** Paper Emissions in 2018-2019 were calculated using Environmental Paper Network Paper Calculator Version 4.0. 2012 Paper Emissions were calculated using Environmental Paper Network Paper Calculator Version 3.2.

ENVIRONMENT	2012	2018	2019
Climate Change and Greenhouse Gas (GHG) Emissions			
Scope 3 GHG Emissions (Metric tons of CO ₂ e)- Supply Chain and Transport (Continued)			
Capital Goods			
Total lifecycle furniture emissions	N/A	1,283	989
Fuel-and-energy related activities			
Electricity transmission/distribution loss emissions	N/A	1,868	1,708
Upstream Transportation and Distribution			
Shipping emissions	N/A	133	195
Armored vehicle emissions	N/A	59	69
Waste Generated in Operations			
Scope 3 lifecycle emissions associated with landfilled mixed solid waste	2,052	514	505
Business Travel			
Employee business travel by air and car	4,431	3,453	3,515
Emissions intensity- Employee business travel emissions (metric tons of CO ₂ e) per full-time employee (FTE)	N/A	0.44	0.46
Employee Commuting			
Employee commuting emissions	N/A	25,544	25,837
Downstream Leased Assets			
<i>Total subleased emissions</i>	1,257	1,027	629
Subleased metered electricity	726	261	397
Subleased estimated electricity	423	560	12
Subleased natural gas	81	25	71
Subleased steam	28	0	0
Subleased estimated heat-natural gas	N/A	6	0
Subleased corporate jet emissions	N/A	175	149
Other Downstream			
CBRE Business Travel on Comerica Account	N/A	181	171

ENVIRONMENT	2012	2018	2019
Real Estate Square Footage (Square feet at year end)			
Total metered and unmetered square feet at year end	5,545,466	4,401,337	4,350,468
Total subleased square feet at year end	137,637	51,522	63,375
Real Estate Square Footage (Average square feet over four quarters)			
Average metered and unmetered square feet over four quarters	N/A	4,547,160	4,370,816
Average subleased square feet over four quarters	N/A	115,654	66,099
Environmental Resource Management			
Waste			
Landfilled employee waste (short tons)	2,086	1,480	1,455
Percent Cumulative reduction from 2012 base year (Goal=20 percent Landfilled Waste Reduction by 2020)	N/A	29.1	30.3
Total recycled/re-purposed waste (short tons)	3,372	2,376	2,184
Recycled office paper	3,108	2,148	1,928
Recycled/repurposed electronic equipment	149	48	87
Recycled operational waste	110	156	154
Recycled cardboard bailer waste	N/A	6	3
Recycled pallets	N/A	12	10
Universal waste	3	5	2
Total waste diverted from landfill (Percent)- including recycled office paper, recycled/repurposed electronic equipment, recycled employee waste, and universal waste	62	62	60
Paper			
Total paper consumption (tons)	1,266	695	602
Office copy paper consumption (tons)	560	289	245
Other office paper consumption (tons)	618	361	336
Marketing paper consumption (tons)	88	45	22
Paper Intensity metrics (pounds consumed per FTE)			
Total office paper (office copy and other office paper) consumption in pounds per FTE	N/A	166.88	152.72
Office copy paper consumption in pounds per FTE	N/A	74.12	64.38
Percent Cumulative reduction from 2012 base year (Goal=50 percent Office Copy Paper Reduction by 2020)	N/A	48.5	56.3

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GRI Standard	Disclosure	Is this topic material?	Link or Direct Comments	Omission
Materials		Yes		
GRI 301: Materials	301-1 Materials used by weight or volume	Yes	Key Metrics Table	—
	301-2 Recycled input materials used	Yes	Key Metrics Table , 2018 CSR Report (Page 72)	—
	301-3 Reclaimed products and their packaging materials	Yes	Key Metrics Table , 2018 CSR Report (Page 72)	—
Energy		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment , Energy & Emissions	—
	103-2 The management approach and its components	Yes	Energy & Emissions , Environmental Policy Statement , 2018 CSR Report (Page 68-70)	—
	103-3 Evaluation of the management approach	Yes	Energy & Emissions , 2018 CSR Report (Page 68-70)	—
GRI 302: Energy	302-1 Energy consumption within the organization	Yes	Energy & Emissions , Key Metrics Table , Most Recent CDP Response	—
	302-2 Energy consumption outside of the organization	Yes	Energy & Emissions , Key Metrics Table , Most Recent CDP Response	—
	302-3 Energy intensity	Yes	Energy & Emissions , Key Metrics Table , Most Recent CDP Response	—
	302-4 Reduction of energy consumption	Yes	Energy & Emissions , Key Metrics Table , Most Recent CDP Response , 2018 CSR Report (Pages 68-70)	—
	302-5 Reductions in energy requirements of products and services	Yes	Energy & Emissions , Key Metrics Table , Most Recent CDP Response , 2018 CSR Report (Pages 68-70)	—
Water		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment , Environmental Resource Management , 2018 CSR Report (Pages 74-75)	—
	103-2 The management approach and its components	Yes	Environmental Resource Management , 2018 CSR Report (Pages 74-75)	—
	103-3 Evaluation of the management approach	Yes	Environmental Resource Management , 2018 CSR Report (Pages 74-75) , Key Metrics Table	—
GRI 303: Water and Effluents 2018	303-1 Water withdrawal by source	Yes	Key Metrics Table	Although we have a couple wells and ponds in Michigan for irrigation purposes, the only water metric that we are able to measure is the water which is sourced through local utilities and is tracked through the utility water meter.

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GRI Standard	Disclosure	Is this topic material?	Link or Direct Comments	Omission
Biodiversity		No		
Emissions		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment, Energy & Emissions	—
	103-2 The management approach and its components	Yes	Energy & Emissions, Environmental Policy Statement, 2018 CSR Report (Pages 70-72)	—
	103-3 Evaluation of the management approach	Yes	Energy & Emissions, 2018 CSR Report (Pages 70-72)	—
GRI 305: Emissions	305-1 Direct (Scope 1) GHG emissions	Yes	Energy & Emissions, Key Metrics Table, Most Recent CDP Response	—
	305-2 Energy indirect (Scope 2) GHG emissions	Yes	Energy & Emissions, Key Metrics Table, Most Recent CDP Response	—
	305-3 Other indirect (Scope 3) GHG emissions	Yes	Energy & Emissions, Key Metrics Table, Most Recent CDP Response	—
	305-4 GHG emissions intensity	Yes	Energy & Emissions, Key Metrics Table, Most Recent CDP Response	—
	305-5 Reduction of GHG emissions	Yes	Energy & Emissions, Key Metrics Table, Most Recent CDP Response	—
Effluents and Waste		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment, Environmental Resource Management	—
	103-2 The management approach and its components	Yes	Environmental Resource Management, Environmental Policy Statement	—
	103-3 Evaluation of the management approach	Yes	Environmental Resource Management, Environmental Policy Statement	—
GRI 306: Effluents and Waste	306-2 Waste by type and disposal method	Yes	Environmental Resource Management, Environmental Policy Statement, Key Metrics Table	—
Environmental Compliance		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment, Environmental Risk Management (2018 CSR Report, Page 78); part of Business Risk Management	—
	103-2 The management approach and its components	Yes	Environmental Risk Management (2018 CSR Report, Page 78)	—
	103-3 Evaluation of the management approach	Yes	Environmental Risk Management (2018 CSR Report, Page 78)	—
GRI 307: Environmental Compliance	307-1 Non-compliance with environmental laws and regulations	Yes	There were no significant fines or monetary sanctions associated with non-compliance of environmental laws and regulations in 2019 that would have a material adverse effect on our operations or ability to perform the services we offer.	—
Supplier Environmental Assessment		No		
GRI 308: Supplier Environmental Assessment	308-1 New suppliers that were screened using environmental criteria	No	Key Metrics Table	—

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GRI Standard	Disclosure	Is this topic material?	Link or Direct Comments	Omission
GRI 400 Social Standards Series				
Employment		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment, Talent Attraction, Development & Retention, Employee Engagement, Diversity & Inclusion Report	—
	103-2 The management approach and its components	Yes	Talent Attraction, Development & Retention, Employee Engagement, Diversity & Inclusion Report	—
	103-3 Evaluation of the management approach	Yes	Talent Attraction, Development & Retention, Employee Engagement, Diversity & Inclusion Report	—
GRI 401: Employment	401-1 New employee hires and employee turnover	Yes	Talent Attraction, Development & Retention, Employee Engagement, Key Metrics- New Hires, Key Metrics-Turnover	—
	401-2 Benefits provided to full-time employees that are not provided to temporary or part-time employees	Yes	Talent Attraction, Development & Retention, 2018 CSR Report (Page 50), Key Metrics Table	—
Labor/Management Relations		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment, Colleagues	—
	103-2 The management approach and its components	Yes	Colleagues, Talent Attraction, Development & Retention, 2018 CSR Report (Pages 45-51)	—
	103-3 Evaluation of the management approach	Yes	Colleagues, Talent Attraction, Development & Retention, Employee Engagement, 2018 CSR Report (Pages 45-51), Key Metrics Table	—
GRI 402: Labor/Management Relations	402-1 Minimum notice periods regarding operational changes	Yes	The minimum notice period for significant operational changes is 45 days (6.4 weeks) unless the change triggers the provisions of the Worker Adjustment and Retraining Notification (WARN) act in which case a minimum of 60 days notice is provided.	—
Occupational Health and Safety		No		
Training and Education		Yes		
GRI 103: Management Approach	103-1 Explanation of the material topic and its Boundary	Yes	ESG Impact Assessment, Talent Attraction, Development & Retention	—
	103-2 The management approach and its components	Yes	Talent Attraction, Development & Retention, Diversity Progress Report, 2018 CSR Report (Pages 46-51)	—
	103-3 Evaluation of the management approach	Yes	Talent Attraction, Development & Retention, Diversity Progress Report, 2018 CSR Report (Pages 46-51)	—
GRI 404: Training and Education	404-1 Average hours of training per year per employee	Yes	Key Metrics Table	—
	404-2 Programs for upgrading employee skills and transition assistance programs	Yes	Talent Attraction, Development & Retention, Key Metrics Table	—
	404-3 Percentage of employees receiving regular performance and career development reviews	Yes	Talent Attraction, Development & Retention, Key Metrics Table	—